A Multimodal Explainable AI Framework for Depression Detection using PHQ-9 and Eye-Based Features

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CERTIFICATE

This is to certify that this project report titled **"A Multimodal Explainable AI Framework for Depression Detection using PHQ-9 and Eye-Based Features"** is a bonafide record of work done by **ALLA HANVITHA SAI (228W1A1202), NEELAPALA MANVITHA (228W1A1241)** under my guidance and supervision is submitted in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Information Technology, **V.R. Siddhartha Engineering College** (Autonomous under JNTUK) during the year 2025-26.

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PROJECT SUMMARY

S.No	Item	Description
1	**Project Title**	**A Multimodal Explainable AI Framework for Depression Detection using PHQ-9 and Eye-Based Features**
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3	**Name of The Guide**	**Dr. G SrinivasaRao**
4	**Research Group**	**Artificial Intelligence and Machine Learning**
5	**Application Area**	**Mental Health Analysis**
6	**Aim of the Project**	**To develop a multimodal depression detection system by integrating PHQ-9 questionnaire data and eye-based image features using machine learning, deep learning, and explainable AI techniques to improve accuracy and interpretability.**
7	**Project Outcomes**	• **Predicts depression severity using multimodal data (PHQ-9 + eye images).** • **Improves prediction accuracy using fusion of Random Forest and Vision Transformer models.** • **Provides explainability using SHAP and attention maps for better interpretation of results.**

Student Signatures

1.

2.

Signature of the Guide

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ABSTRACT

Depression is a critical mental health disorder that requires early and accurate detection to prevent severe consequences. Traditional methods primarily depend on self-reported questionnaires, which may be subjective and lack reliability. To overcome these limitations, this project proposes a multimodal approach for depression detection by integrating both questionnaire-based and image-based data using advanced artificial intelligence techniques.

In this system, Patient Health Questionnaire-9 survey responses are utilized as structured input data, while eye images are used as visual input to capture physiological features related to mental health. The questionnaire data is processed using the Random Forest algorithm, which effectively models relationships between responses and depression severity levels. Simultaneously, eye images are analyzed using a Vision Transformer model, which extracts meaningful features by dividing images into patches and applying attention mechanisms. The outputs from both models are combined using a decision- level weighted fusion technique to improve prediction accuracy and robustness.

Furthermore, the system incorporates Explainable Artificial Intelligence techniques to enhance interpretability. SHapley Additive Explanations is used to determine the importance of each PHQ-9 feature while attention map extraction highlights significant regions in eye images, particularly the pupil area
contributing to the final prediction. Experimental results demonstrate that the multimodal fusion model outperforms individual models in terms of accuracy and reliability.

The proposed system provides an efficient, interpretable, and scalable solution for depression detection, making it suitable for real-world healthcare applications and early mental health screening.

Keywords: Depression Detection, Multimodal Learning, PHQ-9, Vision Transformer, Random Forest, Explainable AI.

CHAPTER-1 INTRODUCTON

1.1 Origin of the Problem

The past few years have seen mental health disorders and especially depression emerging as one of the most critical health issues in the world, impacting all age groups in a significant manner and decreasing their quality of life. Depression does not only impact on the emotional state but also on the cognitive abilities, productivity and social interaction. Although its prevalence is on the rise, many cases are going undiagnosed because of lack of awareness, social stigma and inaccessibility of mental healthcare services. Conventional approaches to depression diagnosis mainly involve clinical interviews and self- report questionnaires, e.g. PHQ-9. These techniques are highly subjective, although they are common and clinically proven and require the honesty and emotional awareness of an individual. This tends to produce biased or inconsistent outcomes, which are hard to detect at the initial stages [1], [2].

As Artificial Intelligence (AI) progresses, scientists have investigated automated processes of mental health detection based on machine learning and deep learning. Recent research evidences that AI models can be effective to analyze behavioral and physiological data to detect depression [1], [4]. Among other physiological indices, the pupil dilation has become a dependable measure of the emotional and cognitive conditions, since it is a symptom of the neurological processes behind [8], [15].

The majority of available systems, however, depend on one modality (questionnaire information or physiological measurements), making them less accurate and reliable. Also, most AI-based systems are black-box models, which are not interpretable and decrease user trust [3], [4]. In order to eliminate these shortcomings, this project has suggested A Multimodal Explainable AI Approach to Depression Detection based on Pupil-based Features and PHQ-9 Responses. It combines subjective (questionnaire-based) and objective (eye-based) data which allows to reliably detect depression and interpret it.

1.2 Background and basic definitions.

Mental health is a state of psychological, emotional, and social health. Depression is a mental condition that is common and is typified by an individual being regularly sad, uninterested in activities, experiencing fatigue and displaying impaired brain functions. The key to effective treatment and prevention of the severe consequences is early detection.

PHQ-9 (Patient Health Questionnaire-9) is a common tool of depression severity assessment that is standardized. It is a questionnaire composed of nine questions, which are rated between 0 and 3, and can be classified into categories such as minimal, mild, moderate and severe depression. Although it works, it is not objective and can not necessarily be relevant to the real mental condition of a person.

Pupil dilation is an autonomic nervous system response to emotional stimuli, cognitive load, and neurological activity, which varies. It has been demonstrated that depressed individuals have distortions in their pupil reactions and pupil-based analysis is a useful non-invasive technique of detecting mental health conditions [8], [9], [18].

Machine Learning is extensively applied to classification in the healthcare sector. Random Forest algorithms can be used in structured data, like the answers to questionnaires, and make accurate predictions [1], [3]. Image analysis tasks are performed with deep learning models like Convolutional Neural Networks (CNN) and Vision Transformers, which allow extracting meaningful information about eye images [11], [12].

Vision Transformer (ViT) is a recent deep learning model that processes images through patching and using attention mechanisms. This enables the model to record the worldly dependencies and minor differences in the features of the eye and thus it is applicable in the analysis of the patterns based on the pupil.

Multimodal learning is a conglomeration of various data sources to enhance prediction accuracy. PHQ- 9 responses give subjective information in this project whereas the features based on pupils give objective physiological data. Such a mixture increases the reliability of the system, and eliminates constraints related to single-modality approaches [1], [5].

Explainable AI (XAI) is concerned with increasing the transparency and interpretability of machine learning models. SHapley Additive Explanations can be used to determine the importance of features in random forest models and attention maps in vision transformers can be used to determine regions of importance in eye images. This enhances user confidence and allows to comprehend model decisions better [3], [4].

1.3 Problem Statement

The rise of mental illnesses especially depression has become a world issue especially to people of all ages and socio-economic status. On top of affecting emotional health, depression also results in limited productivity, poor cognitive performance and serious health issues in case it is not managed at an early age. Although clinical diagnostic methods are available, a considerable amount of the cases are not diagnosed or are late diagnosed, which restricts the quality of treatment. Conventional means of depression detection are self-reported forms of questionnaires like the Patient Health Questionnaire-9 and mental health professionals interviews. Even though these approaches are commonly accepted and proved clinically, they have certain limitations including subjectivity, response bias, and reliance on the willingness of the individual to report the personal emotions in a correct manner. People in most instances might either underreport or distort their symptoms, either as a result of social stigma, ignorance or fear of being judged, hence resulting to inaccurate diagnosis and late intervention [1], [2].

To address these drawbacks, in recent years, the application of Artificial Intelligence methods to automated mental health detection has been investigated. Machine learning and deep learning system

have demonstrated potential outcomes in behavioral and physiological data analysis in order to determine patterns related to depression [1], [4]. Among other physiological markers, the pupil dilation has become a good and non-invasive biomarker in that it is a measure of the underlying cognitive and emotional activities [8], [15]. Eye based analysis offers objective data on the mental status of a person with no active involvement and hence is appropriate in real time monitoring.

Nevertheless, current AI-driven depression detection systems have a number of issues. The majority of the existing methods are based on one type of analysis, either a questionnaire-based data or a physiological signal. Questionnaire-based systems can be considered lacking objectivity and those relying on physiological data alone cannot take into consideration a clinical setting, which leads to low accuracy and reliability [5]. There are also numerous state-of-the-art deep learning models that are black-box and highly accurate but have no interpretability. This complicates users and health care professionals to comprehend the rationale behind predictions thus diminishing trust and restricting real world application [3], [4]. Moreover, some of the current approaches involve a dedicated hardware, like eye-tracking machines, or VR systems, making them costly and impossible to scale [6], [7]. It is required that a solution is not only accurate but also accessible, affordable and simple to implement using devices that are commonly available like webcams.

Hence, the main issue, which is dealt with in this project, is the creation of a strong, multi-modal, and explainable system of detecting depression, which will jump over the shortcomings of current solutions. To enhance the accuracy of prediction, the system is to combine subjective and objective sources of data, i.e., Patient Health Questionnaire-9 responses and pupil-based attributes derived with the help of eye images. It must also use Explainable AI practices to improve transparency and make meaningful insights on model decisions.

The proposed solution aims to address these challenges by combining machine learning and deep learning models in a multimodal framework. The system aims to give an accurate, reliable, and interpretable early detection of depression through the combination of a Random Forest model to analyze questionnaire data and a Vision Transformer to analyze eye images with explainability methods, such as SHapley Additive Explanations and attention maps. This will provide healthcare professionals with a higher level of decision-making support and provide real-world applications with accessibility and scalability.

1.3.1 Objectives

- To create a multimodal depression detector, combining Patient Health Questionnaire-9 data with pupil- based eye image analysis.

- To apply machine learning and deep learning models (Random Forest and Vision Transformer) to the correct classification of depression levels.

- To include Explainable AI models like SHapley Additive Explanations and attention maps to enhance interpretability and transparency of the system.

1.3.2 Outcomes

- Real-time, non-invasive, physiological and clinical based, depression detection system.

- Better accuracy and reliability in prediction compared to systems that use single-modality data.

- User trust and greater model transparency through explainable results, which can be achieved through Explainable AI techniques.

1.4 Related Applications

The multimodal explainable AI method of depression detection proposed system can find a broad range of applications in different fields. The system can effectively be used to assess mental health early in life by combining the answers of pupil-based physiological characteristics and Patient Health Questionnaire-9, offering a non-invasive and reliable solution.

1.4.1 Healthcare Systems

The system can also help medical professionals to diagnose and monitor the progress of depression in its initial stages in the healthcare sector. Conventional clinical practices may be time consuming and based on subjective assessments. The given system offers objective data, which is based on eye-analysis and makes the diagnosis quicker and more correct. Moreover, Explainable AI techniques can assist clinicians to comprehend the logic behind predictions, enhancing trust and usefulness in medical practice.

1.4.2 Educational Institutions

Academic pressures and personal challenges often make students experience stress, anxiety, and depression. The proposed system may serve as a screening device to observe the mental health of the students and detect the early symptoms of mental disturbance. Early identification enables timely action by the counselors or educators, which helps to avoid a serious mental illness and enhance student performance.

1.4.3 Workplace Mental Health Monitoring

Mental health among the employees is a very critical issue that determines the productivity, efficiency, and job satisfaction. This system is able to help organizations track the level of stress and also identify early signs of depression in employees. It is through the insights made that companies can adopt wellness programs, counseling sessions and preventive measures to ensure that the work environment is healthy.

1.4.4 Remote Healthcare and Telemedicine.

As the application of digital health systems becomes more widespread, the suggested system may be applied in remote mental health evaluation. A webcam-based program will enable users to go through the screening process without going to hospitals. This is especially helpful to people in rural or underserved communities where mental health practitioners are scarce.

the proposed system will help fill the gap between the existing method of mental health assessment and the current AI-based solutions, as depression detection will be more accessible, efficient, and reliable.

CHAPTER-2 LITERATURE REVIEW

2.1 Description of Existing Systems

2.1.1 Title: Effective Depression Detection and Interpretation: Integrating Machine Learning, Deep Learning, Language Models, and Explainable AI

Authors: Gazi Hasan Al Masud, Rejaul Islam Shanto, Ishmam Sakin, and Muhammad Rafsan Kabir

Year of Publication: 2025

The present research suggests a holistic method of detection of depression via a combination of machine learning, deep learning, natural language processing, and explainable AI. It employs a survey sample of about 1600 students of Bangladeshi university with demographic, lifestyle, and psychological variables. Different algorithms like Logistic Regression, Decision Tree, Random Forest, SVM, KNN, AdaBoost, and XGBoost were executed, and the best algorithm was the Random Forest (with an accuracy of 91.1%). Other deep learning models such as MLP and CNN were also utilized but demonstrated slightly less performance. Also, BERT, DistilBERT and RoBERTa language models were used, with RoBERTa showing a high recall. Predictions were interpreted through explainable AI algorithms such as SHAP and LIME to find the most important features such as anxiety and loneliness. Nevertheless, the study has weaknesses in terms of the size of the dataset, bias of the population, and the use of synthetic data, which can impact the generalizability. In general, the system proves the effectiveness of integrating various AI methods with explainability in enhancing the performance of depression detection [1].

2.1.2 **Title:** Artificial Intelligence-Based Depression Detection

Authors: Gabor Kiss and Patrik Viktor

Year of Publication: 2026

This paper provides a research-based artificial intelligence system of depression detection based on physiological measurements of eye behavior, such as pupillary reactions and eye movement. This system is made up of iris-based identification module and a depression detection module where the eye-tracking data (pupil dilation, fixation duration, and gaze patterns) is analyzed. It utilizes a hybrid deep learning model, which is a combination of CNN and LSTM with CNN extracting spatial features and LSTM capturing the pattern over time. The model obtained the accuracy of about 89% and the high AUC value indicating that it has been working well. The paper also presents an individualized baseline method to enhance the accuracy of detection. Nevertheless, the system involves specific eye- tracking devices and can be influenced by other factors like light, exhaustion and stress as well as the privacy issues. In general, the paper shows that AI systems based on physiological signals have the potential to facilitate accurate detection of depression in real-time [2].

2.1.3 **Title:** Explainable Artificial Intelligence Models for Predicting Depression Based on Polysomnographic Phenotypes

Authors: Doljinsuren Enkhbayar, Jaehoon Ko, Somin Oh, Rumana Ferdushi, Jaesoo Kim, Jaehong Key, and Erdenebayar Urtnasan

Year of Publication: 2025

This paper suggests a predictive AI model to explain depression based on polysomnographic (PSG) sleep recordings. The data is a product of the BESTAIR study where 153 participants were studied based on clinical and physiological parameters, including sleep efficiency, apnea index, and demographic information. Several machine learning models have been used such as the Random Forest, XGBoost, LightGBM, and CatBoost with the best performance of approximately 85% accuracy with the use of the random forest and the catboost. The research combines SHAP to explain, and the following traits of sleep efficiency and anxiety are crucial factors in depression. Findings indicate that there is a close correlation between sleep disturbances and depression. The system is however constrained by limited data set and reliance on PSG equipment making it difficult to use it on a large scale. In general, the experiment shows that PSG data combined with explainable AI can be an effective and interpretable method of detecting depression [3].

2.1.4 **Title:** Explainable AI-Driven Depression Detection from Social Media Using Natural Language Processing and Black Box Machine Learning Models

Authors: S. Hameed, M. Nauman, N. Akhtar, M. A. B. Fayyaz, and R. Nawaz

Year of Publication: 2025

This paper offers a clearable AI-driven model to identify depression on social media data using natural language processing and machine learning algorithms. The dataset is large (1.6 million tweets), with preprocessing of text and extraction of features performed by means of TF-IDF, Bag of Words, N- grams, and GloVe embeddings. There are different models such as SVM, Random Forest, XGBoost, and ANN and the SVM and GloVe based models perform best with an accuracy of about 88%. The analysis combines LIME to enable explainability, determining the key words that affect the predictions and enhance transparency. Nonetheless, the system is constrained by the noisy and ambiguous social media data, labeling based on keywords, and privacy issues. In general, the paper indicates that a combination of NLP, machine learning, and explainable AI represent a powerful and scalable strategy to identify depression [4].

2.1.5 Title: PupilSense: Detection of Depressive Episodes Through Pupillary Response in the Wild

Authors: Rahul Islam and Sang Won Bae

Year of Publication: 2024

The study presents a mobile-based system named PupilSense used to detect depressive episodes in real-life situations based on detecting pupillary response. The system takes pictures of eyes with the front cameras of smartphones when it is in use and identifies the things like the pupil-to-iris ratio (PIR) with the help of a deep learning model, which is built on Detectron2. The PIR values are computed and then used to compute the statistical features that are used to classify with machine learning models like Decision Trees and SMOTE to deal with the class imbalance. The system is tested on 25 participants and has an accuracy of about 76%. It has the main advantage of passive continuous monitoring without the need of additional hardware. The performance is however influenced by the lighting conditions, small size of data set used and privacy concerns. In general, the research has shown that pupillary analysis via mobile can be a promising method of detecting depression in real-time [5].

2.1.6 **Title:** Method and Device for Automatically Detecting Early Depression Using Multimodal Physiological Signals

Authors: M. Li, Y. Chen, Z. Lu, F. Ding, and B. Hu

Year of Publication: 2024

The proposed study is based on the high-performance multimodal paradigm of detecting early depression with the use of physiological indicators of this condition, including pupil wave measurements and pulse rate variability (PRV) in an emotional and non-emotional condition. The system uses a hybrid deep learning model that has BiTCN to extract temporal features, BiLSTM to model in a sequence, and attention to prioritize significant emotional and multimodal features. An ensemble learning approach is used for final classification, along with data augmentation to improve generalization. The results show that incorporating emotional signals significantly improves detection accuracy, achieving state-of-the-art performance. Another new device added to the system is the Automatic Detection of Early Depression Device (ADED) to be used in real-time. Nevertheless, it has high computational complexity, and needs specialized hardware which restricts large-scale implementation. In general, the paper illustrates that multimodal physiological cues plus deep learning and attention algorithms can be successfully used to detect depression [6].

2.2 Summary of Literature Study

Ref.no.	Publication Details	Datasets	Algorithms	Limitations
1.	Masud et al., *"Effective Depression Detection and Interpretation: Integrating Machine Learning, Deep Learning, Language Models, and Explainable AI,"* Array, 2025 [1]	Survey dataset (1600 student records)	ML (RF, SVM, DT), DL (MLP, CNN), NLP (BERT, RoBERTa), XAI (SHAP, LIME)	Limited dataset, synthetic data bias
2.	Kiss & Viktor, *"Artificial Intelligence-Based Depression Detection,"* Sensors, 2026 [2]	Eye-tracking & pupillary data	CNN + LSTM	Requires hardware, affected by lighting & fatigue
3.	Enkhbayar et al., *"Explainable Artificial Intelligence Models for Predicting Depression Based on Polysomnographic Phenotypes,"* Bioengineering, 2025 [3]	PSG dataset (153 subjects)	RF, XGBoost, LightGBM, CatBoost + SHAP	Requires PSG equipment, small dataset
4.	Hameed et al., *"Explainable AI-Driven Depression Detection from Social Media Using NLP and ML Models,"* Frontiers in Artificial Intelligence, 2025 [4]	Social media (1.6M tweets)	NLP (TF-IDF, GloVe), ML (SVM, RF, XGBoost), XAI (LIME)	Noisy data, privacy issues

5.	Islam & Bae, *"PupilSense: Detection of Depressive Episodes Through Pupillary Response in the Wild,"* Sensors (Mobile Health), 2024 [5]	Smartphone pupil data (25 users)	Detectron2 (DL), Decision Tree, SMOTE	Small dataset, lighting issues
6.	Li et al., *"Method and Device for Automatically Detecting Early Depression Using Multimodal Physiological Signals,"* IEEE Publication, 2024 [6]	Pupil waves + PRV signals	BiTCN, BiLSTM, Attention, Ensemble	Complex model, hardware dependency
7.	Zheng et al., *"Diagnosing and Tracking Depression Based on Eye Movement in Response to Virtual Reality,"* Frontiers in Psychiatry, 2024 [7]	VR eye movement data	MLP, SVM, XGBoost, RF	Requires VR setup
8.	Spaeth et al., *"Are Depressive Symptoms Linked toa Reduced Pupillary Response to Novel Positive Information?"* Journal of Affective Disorders, 2024 [8]	Eye-tracking (34 participants)	BADE task, statistical analysis	Small sample size
9.	Alshanskaia et al., *"Stress Detection Using Cognitive Load from Pupil Dilation,"* Neural Computing and Applications, 2024 [9]	Eye-tracking dataset (pupil dilation under cognitive tasks)	ML (SVM), statistical analysis, feature extraction (pupil size, variation)	Sensitive to lighting conditions, limited dataset,

2.3 Research Gap

- Existing systems primarily rely on single-modal data, which limits performance. This work proposes a multimodal approach by integrating physiological signals with questionnaire data.

- Many high-performing models operate as black-box systems, reducing trust and interpretability. This study incorporates Explainable AI (XAI) to enhance transparency.

- Current approaches often depend on expensive and specialized hardware, making them impractical for large-scale use. The proposed system focuses on a cost-effective and scalable solution.

CHAPTER-3 Proposed Method

3.1 Design Methodology:

The system proposed is a multimodal framework of depression detection, which combines both visual eye image data and questionnaire-based data, to enhance the accuracy and interpretability of the prediction. The approach is a mix of machine learning and deep learning and explainable artificial intelligence (XAI) to deliver credible and transparent outcomes.

The system receives two kinds of inputs, Patient Health Questionnaire-9 and eye images. Patient Health Questionnaire-9 data set is composed of nine items each with a rating of 0 to 3 based on the intensity of depressive symptoms. These answers are computed to form a cumulative score and to classify people into various severity levels as minimal, mild, moderate, moderately severe and severe depression.

For processing questionnaire data, a Random Forest classifier is used. The reason why this model is selected is because it is very accurate, robust and able to manipulate structured data. It also gives the values of feature importance which can be used to give interpretability.In the case of visual data, a Vision Transformer model is used to process eye pictures. Preprocessing of the images involves resizing the images to 224 224 pixels and placing them in classes. The ViT model extracts deep visual features and learns patterns related to depression from eye characteristics.

A fusion technique is used to combine the outputs of the two models. Probability scores of the Vision Transformer and the Random Forest model are combined in this method to make the final prediction. This multimodal fusion is better in performance than, single models.

Explainable AI techniques are also implemented in the system on several levels. The explainability at the feature level is attained by using SHapley Additive Explanations values of the Random Forest model, which indicates the significance of each questionnaire feature. The explainability on the image level is available on the attention maps offered by the Vision Transformer that detects significant areas in eye images. Moreover, the modality-level explainability demonstrates the input of each model to the eventual forecast

In sum, the offered methodology offers a powerful, explainable and effective approach to detect depression, as it integrates clinical and visual data, solving the shortcomings of current single- modality systems.

3.2 System Architecture Diagram

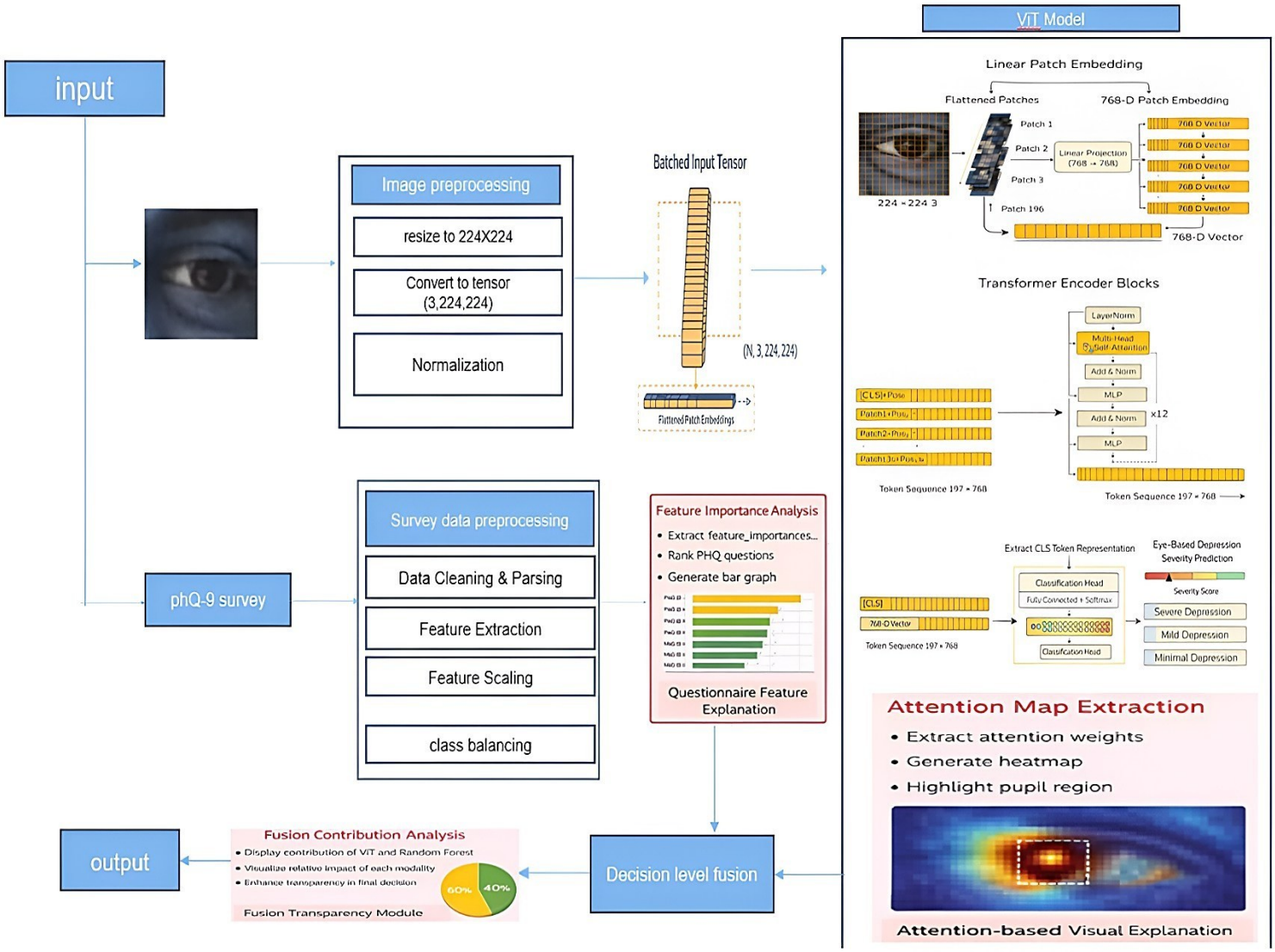


Figure 3.2.1: Architecture Diagram

The proposed system architecture Figure 3.2.1 is a multimodal depression detection framework, combining visual and questionnaire data, to enhance accuracy and interpretability of predictions.

The system takes two types of inputs: eye images and Patient Health Questionnaire-9 survey data. They are fed into different pipes and eventually fused together by a fusion mechanism.

The image processing module is the first module where input eye image is preprocessed by resizing the image to 224 x 224 pixels, converting it into a tensor format (3x224x224), and performing normalization. The resulting processed images are fed into the model as batched input tensors to the Vision Transformer (ViT) model. ViT model subdivides the image into patches, and then it transforms them into embeddings through a linear projection. These embeddings are encoded via transformer encoder blocks comprising of multi-head self-attention and feed-forward layers. The last classification head outputs the prediction of the depression severity using visual features.

At the same time, Patient Health Questionnaire-9 responses in the survey data processing module are preprocessed to clean and parsify the data, extract features, and scale the features. The class balancing methods are also used to manage skewed data. These processed features are then trained to a Random Forest classifier, which classifies the severity of depression, given questionnaire responses.

The system incorporates Explainable AI in order to improve interpretability. Attention maps are also obtained in the Vision Transformer and are used in the image pipeline to focus on important areas within the eye image, especially the pupil area. The feature importance analysis is carried out in the survey pipeline so that the most significant questionnaire features that affect the prediction are found.

The products of the two models are fused with a decision-level fusion mechanism. At this step, the probability scores of the Vision Transformer and Random Forest models are combined to come up with the final prediction. Also, a fusion contribution analysis module is computed to determine the percentage contribution of each modality which enhances transparency in decision-making.

Lastly, the system generates the output that consists of the predicted depression severity and visual explanations (attention maps) as well as feature-based explanations. This design guarantees enhanced precision, interpretability, and relevance of the depression detection system.

3.3 Description of Algorithms

The proposed system is developed by extending the work carried out in Mini-1 and Mini-2 projects. It integrates multiple algorithms including Multilayer Perceptron, Random Forest , Vision Transformer, Weighted Fusion, and Explainable AI techniques to achieve accurate and interpretable depression detection.

3.3.1 Multilayer Perceptron (MLP) – Mini Project 1

Multi-layer perceptron is a type of feedforward artificial neural network employed in classification. This project uses multilayer perceptron to categorize mental health conditions, including Normal, Stress and Depression, using pupillometric characteristics. The model acquires complicated trends using input data by running it through various layers of neurons.

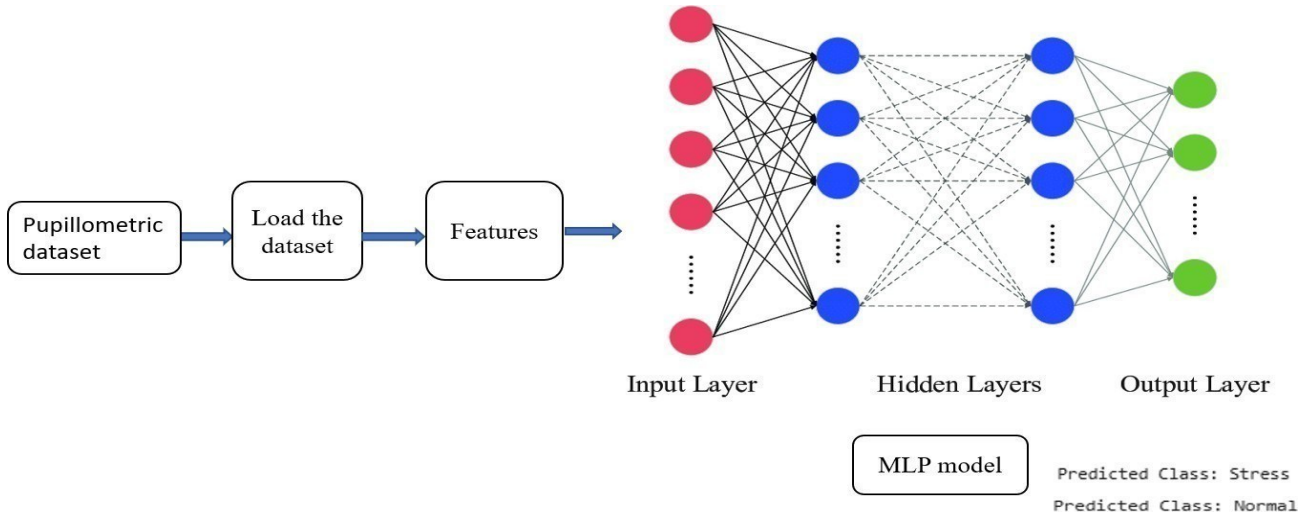


Figure 3.3.1: Multilayer Perceptron Architecture for Mental Health Classification Diagram

The figure 3.3.1 above shows the architecture of the Multilayer Perceptron model to be used in this project. Pupillometric features are fed through the input layer to the hidden layers where they learn the intricate patterns.

The final classification of mental health conditions comes out of the output layer. The operation of multilayer perceptron starts with the input layer where the feature vectors of the dataset are presented to the model. The inputs are a weighted sum of each neuron with a bias term which can be mathematically written as:

$$z = w_i x_i + b \qquad (3.1)$$

where x_i represents input features, w_i represents weights, and b is the bias. This value is then passed through a non-linear activation function such as ReLU, enabling the model to capture complex relationships in the data.

The processed values are propagated with one or more hidden layers and the network learns hierarchical feature representations. Only in the output layer, a Softmax function is used to produce probability scores of each class, and the one with the highest probability is chosen as final prediction. Backpropagation is used to train the model, with the difference between the predicted and actual output minimized by changing the weights after each epoch. It is through this that the model can better its performance and be able to classify accurately. MLP works well with non-linear data and it is also reliable in analyzing physiological signals. Nonetheless, it is also limited to one modality, which can decrease accuracy in real-life situations when it comes to mental health conditions being affected by several factors.

3.3.2 Random Forest Algorithm – Mini Project 2 & Major Project

Random Forest is a machine learning algorithm that is an ensemble of a series of decision trees and is trained to create a series of decision trees and then combine the result of the decision trees together to generate a final prediction. It enhances the stability and accuracy of the model, overall, by minimizing overfitting in comparison to a single decision tree.

$$Final\ Output\ =\ mode(Y_1, Y_2, Y_3, \dots, Y_n) \qquad (3.2)$$

The formula above shows that the end result of the prediction will be the class that will be the most common in the output of all decision trees. Random Forest is applied to the PHQ-9 questionnaire data in this system with each question and total score being the input features. The model acquires correlation between these features and varying levels of depression severity. Random Forests are trained on random subsets of the data and features, which provide variety in decision trees. This randomness assists the model to generalize more on the invisible data. Moreover, the Random Forest can easily cope with the missing values and the noisy data.

A major benefit of Random Forest is that it can give scores of feature importance, giving an idea of the contribution each feature makes to the prediction. The values are further used in SHAP analysis to enhance model interpretability and transparency.

3.3.3 Vision Transformer (ViT) – Mini Project 2 & Major Project

Vision Transformer is a neural network that is used to classify images into categories. As opposed to conventional convolutional neural networks, it works with images by breaking them into small patches and establishing relationships between the patches with the help of attention mechanisms.

$$Attention\ Score = (Q \times K) / \sqrt{d} \qquad (3.3)$$

This equation depicts the way this model computes attention scores to enable it pay attention to the most pertinent aspects of the image. This system works with eye images and the first step involves the resizing and preprocessing of the image and it is divided into fixed sized patches. The patches are flattened and transformed into an embedding by a linear projection layer. These embeddings are further subjected to several layers of transformer encoders. Multi-head self-attention and feed-forward networks make up the encoder layers, facilitating the model to extract local and global features of the image.

This is especially applicable in detecting the minor patterns in the pupil area related to depression. Vision Transformer also offers attention maps, which graphically indicate the areas of the image that contribute to the prediction. This renders the model not only accurate but interpretable as well.

3.3.4 Weighted Fusion Technique

The results of the Random Forest and the Vision Transformer models are fused with a decision-level weighted fusion method to improve the overall performance of the prediction. This method combines both the questionnaire based and image based model information.

$$Final\ Prediction = w_1(RF) + w_2(ViT) \qquad (3.4)$$

The formula above is the combination of the outputs of the two models with a set of weights to get the final prediction. The models generate probability scores of the various classes of depression severity. These scores are weighted and added up to produce a single prediction. The system is more reliable by combining the subjective (Patient Health Questionnaire-9 responses) and the objective (eye image features) information. Fusion aids in overcoming the shortcomings of single models.

Although Random Forest is good with structured questionnaire data, Vision Transformer learns visual patterns. They combine to give a stronger and more precise forecast.

3.3.5 SHAP (SHapley Additive Explanations)

Shapley Additive Explanations is an Explainable AI method to explain the predictions of machine learning models by estimating the importance of each input feature by a number. It is founded on the principles of cooperative game theory, in which each feature can be seen as a player that adds to the final prediction.

The end result will be the sum of the contribution made by each feature to the starting base value. SHapley Additive Explanations is used with the Random Forest model in this system, to examine the influence of each feature of the PHQ-9 questionnaire. The questions, including sleep patterns, mood, energy level, and concentration, all have the importance value indicating the role they play in the prediction of the severity of depression.

The SHapley Additive Explanation values are plotted in feature importance plots and bar graphs, which effectively indicate the contribution of various features to the final prediction as positive or negative. This assists to determine the most severe symptoms of depression. The largest benefit of SHAP is that it has both global and local interpretability. Global interpretation is used to gain insight into the general model behavior and local interpretation is used to gain insight in individual predictions.

This enhances transparency, reliability, and trust with the system particularly in sensitive areas such as mental health analysis.

3.3.6 Attention Map Extraction (ViT Explainability)

Attention map extraction is used to provide visual interpretability for the predictions of the Vision Transformer model. It helps in identifying the regions of the input image that contribute the most to the classification outcome.

In the Vision Transformer, the input image is divided into fixed-size patches (e.g., 16 × 16 pixels), and each patch is converted into a vector representation called a patch embedding. The attention mechanism operates on these patch embeddings rather than directly on individual pixels.

The attention weights are computed using the scaled dot-product attention mechanism:

$$Attention(Q, K, V) = Softmax\left(\frac{QK^T}{\sqrt{d_k}}\right) V \qquad (3.5)$$

where Q, K, and Vrepresent the query, key, and value matrices, and $d_k$is the dimension of the key vectors. The term

$$Softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)$$

represents the attention weights, indicating the importance of each image patch.

The attention map is then obtained by projecting these attention weights back onto the spatial arrangement of image patches:

$$Attention\ Map = Reshape(Attention\ Weights)$$

The resulting attention map is further upsampled to match the original image resolution and visualized as a heatmap. These heatmaps are superimposed on the original eye images to highlight the regions where the model focuses while making predictions.

Typically, the highlighted regions correspond to the pupil and surrounding eye areas, which are important physiological indicators for depression detection. This ensures that the model is focusing on meaningful features rather than irrelevant regions.

This method enhances the interpretability of the deep learning model by providing a clear visual representation of its decision-making process. It also helps in validating the model's predictions and increases confidence in the system outputs.

3.4 Description of Datasets, Requirements and Tools

This section describes the datasets used, along with the software and hardware requirements for implementing the proposed multimodal depression detection system.

3.4.1 Dataset Description

In this section, the datasets employed, software and hardware specifications of the proposed multimodal depression detection system would be outlined.

The suggested system makes use of a multimodal dataset which comprised of questionnaire based data and image based data. The first one is the Patient Health Questionnaire-9 questionnaire data set, which includes the answers of the respondents. Patient Health Questionnaire-9 is a set of 9 questions, which are rated out of 0 to 3. A total score is calculated using these responses and it is used to classify people into various levels of depression severity as minimal, mild, moderate, moderately severe and severe.

The data is saved in a CSV file (labels.csv) that has the following fields:

- Participant ID
- Individual responses (Q1–Q9)
- Total Score
- Severity Label

The second one is the eye image dataset that consists of 2824 eye images taken during the assessment procedure. These images are divided into various severity groups in accordance to the Patient Health Questionnarie-9 scores.

Image details:

- Resolution: Resized to 224 × 224 pixels
- Format: JPG
- Organization: Saved in class-wise folder

This structured data of questionnaire and visual image data allows the system to be able to make multimodal analysis

3.4.2 Software Requirements

The system is implemented with the help of the following software tools and libraries:

- Programming Language: Python
- Development Environment: Google Colab

Libraries Used:

- NumPy and Pandas for data preprocessing
- scikit-learn for implementing Random Forest
- PyTorch and Torchvision for Vision Transformer
- Matplotlib for data visualization
- SHAP for explainable AI

These tools facilitate the effective processing of the data, training of the model, and visualization.

3.4.3 Hardware Requirements

The training and testing of the models would need simple hardware resources to run the system:

- Minimum 8 GB RAM

- Standard CPU system (GPU available as an option to train faster)

- Storage datasets and model files storage.

- Webcam (to capture eye image in real time during earlier stages)

3.4.4 Tools and Frameworks

The following tools and frameworks are used in the development process:

- Machine Learning Framework: scikit-learn

- Deep Learning Framework: PyTorch.

- Image Processing: OpenCV / Torchvision.

- SHAP: An explainable AI tool.

- Visualization Tools: Matplotlib

These tools assist in constructing, educating, analyzing and understanding the models in a competent manner.

CHAPTER 4 RESULTS AND OBSERVATIONS

4.1 Stepwise Description of Results

The results of the proposed work are analyzed in three stages: Mini Project 1, Mini Project 2, and Major Project. Each stage shows progressive improvement in performance, robustness, and interpretability.

4.1.1 Mini Project 1 (MLP Model)

In Mini Project 1, a Multilayer Perceptron model was used to classify mental health conditions based on pupil-related features.

Training and validation metrics and classification results are used to assess the performance of the Multilayer Perceptron model. The figures below depict the model performance in training and testing.

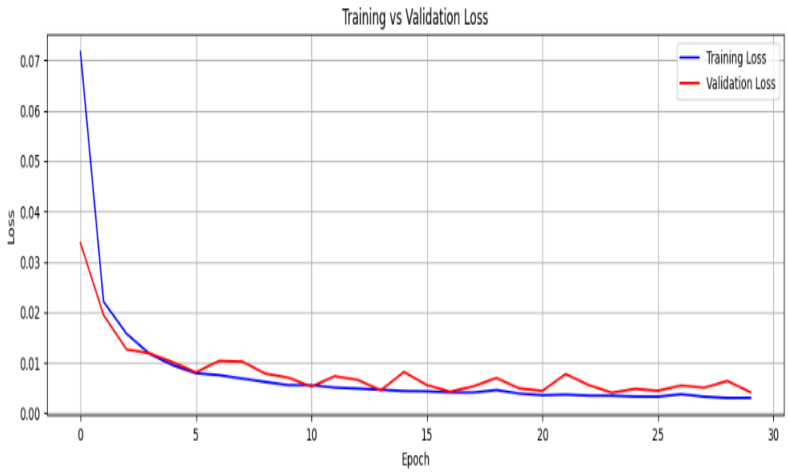


Figure 4.1: Training vs Validation Loss graph

The graph above depicts the change in training and validation loss with the increase in the number of epochs. At first, the loss declines quickly, which means that the model is learning successfully. Training and validation loss approach a very low value, indicating a good convergence, as the epochs increase. The low difference between the two curves shows that the model is not overfitted and can be used to predict unseen data

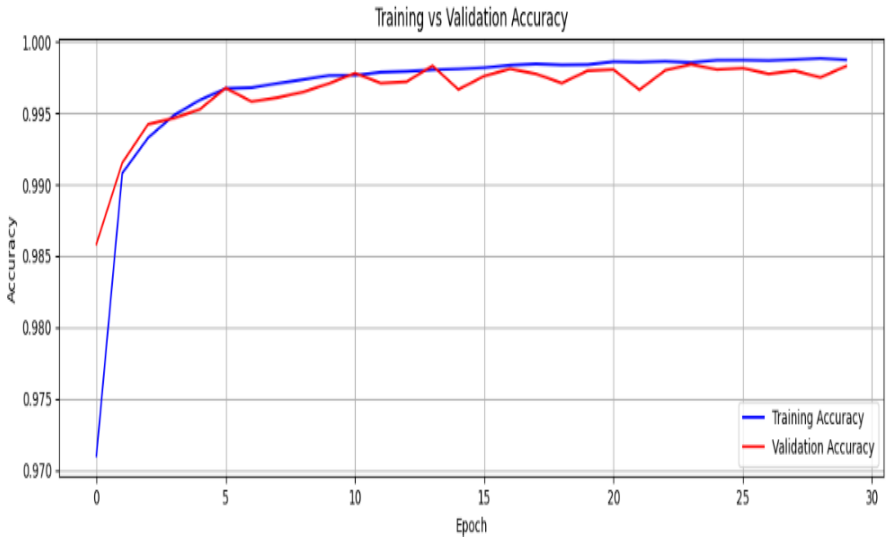


Figure 4.2: Training vs Validation Accuracy

This Figure 4.2 is used to depict the accuracy of the model in both training and validation stages. The accuracy is increasing gradually and is almost 99, meaning that the model is very effective on both training and the validation data. The two curves are very close and this implies that learning is consistent and the models are very reliable.

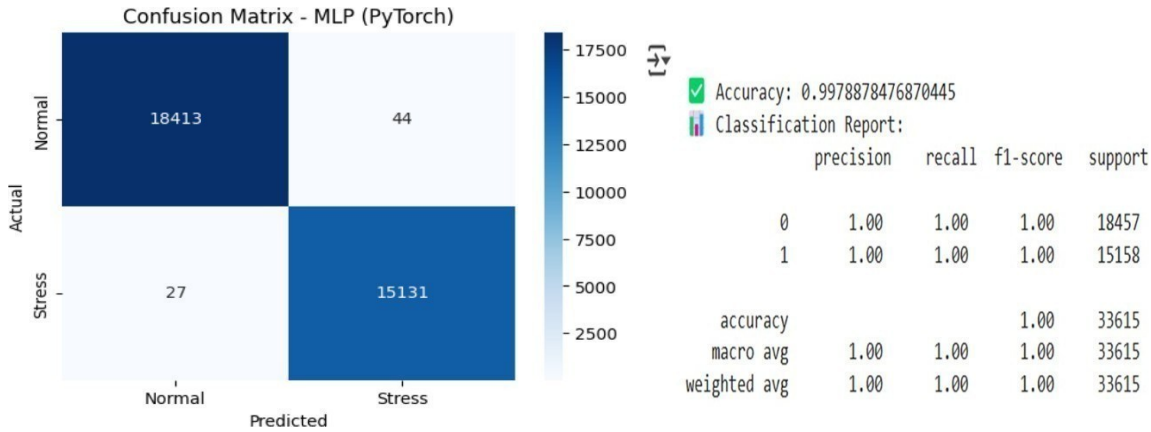


Figure 4.3: Confusion Matrix and Classification Report

The confusion matrix shows the performance of the multilayer perceptron model in terms of classification. It shows that most of the instances are correctly classified into their respective categories (Normal and Stress). Very few misclassifications are made and this implies high precision.

This observation is further supported by the classification report which indicated that the precision and recall and the F1-score are near to 1.00 in both classes. The general accuracy of model is about 99.7 which is great performance of the model in predicting mental health based on pupillometric features.

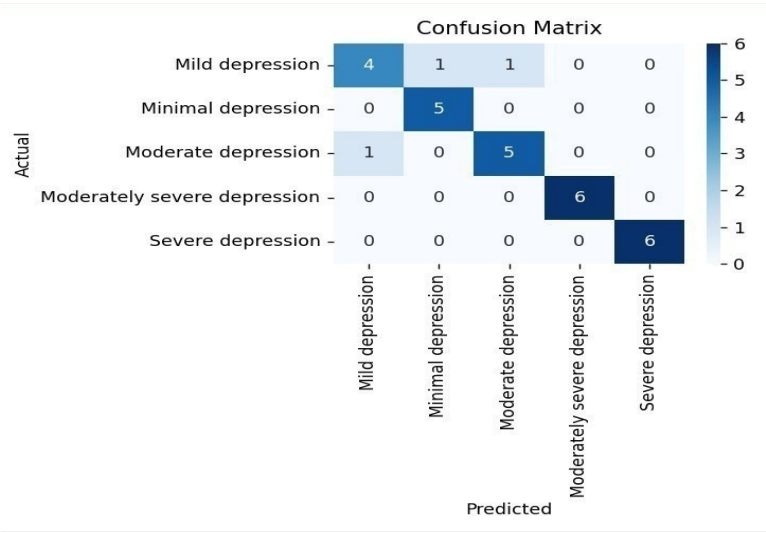
The findings of Mini Project 1 suggest that MLP model can be very effective in the patterns of capturing features based on the eyes and give correct classification with minimum error.

4.1.2 Mini Project 2 (Random Forest and Vision Transformer)

In Mini Project 2, a multimodal approach was introduced using Random Forest for questionnaire data and Vision Transformer for image-based analysis.

Random Forest Results

Random Forest model performance is measured in terms of confusion matrix, classification report and cross-validation accuracy. These findings indicate that the model is effective in the prediction of the level of depression on the basis of PHQ-9 questionnaire responses.




```
Accuracy: 0.896551724137931

Classification Report:
                          precision    recall  f1-score   support

           Mild depression       0.80      0.67      0.73         6
       Minimal depression        0.83      1.00      0.91         5
       Moderate depression       0.83      0.83      0.83         6
Moderately severe depression     1.00      1.00      1.00         6
       Severe depression         1.00      1.00      1.00         6

                accuracy                           0.90        29
               macro avg       0.89      0.90      0.89        29
            weighted avg       0.90      0.90      0.89        29

Cross-Validation Scores: [0.89655172 0.79310345 0.86206897 0.86206897 1.
Mean CV Accuracy: 0.8827586206896552
```

Figure 4.4: Confusion Matrix and Classification Report (Random Forest)

Figure 4.4 shows the performance of the Random Forest model in classifying various levels of depression mild, minimal, moderate, moderately severe, and severe depression. The majority of the predictions are in the diagonal of the matrix which implies proper classification of the instances. There are a couple of misclassifications, especially between closely related classes like mild and moderate depression, as is anticipated because of the overlapping symptoms.

Nevertheless, the model has been shown to be exceptionally good in detecting higher levels of severity like moderately severe and severe depression. The classification report also confirms the performance with the values of precision, recall, and F1-score being close to 1.00 on most of the classes. There is a high predictive performance of the model with a general accuracy of about 89.65. Balanced performance in all classes is also reflected in the macro and weighted averages.

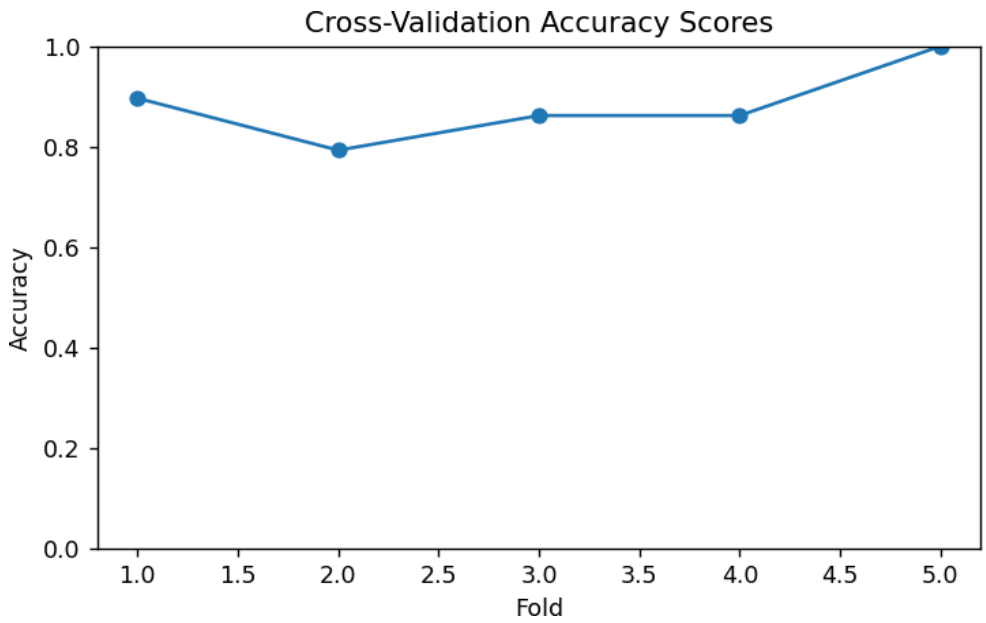


Figure 4.5: The cross-validation accuracy scores

Figure 4.5 indicates how the Random Forest model performs in terms of accuracy in each of the folds. The values of accuracy are high in all folds and the approximate range is between 0.79 and 1.00. The relative consistency in the accuracy suggests that the model is not very sensitive to the training data. An average cross-validation accuracy of about 0.88 indicates that the model is generalized to previously unknown data.

Vision Transformer Results

Vision Transformer model is tested with the help of confusion matrix, classification report, and training-validation accuracy graphs. These results highlight the model's ability to classify depression severity levels based on eye image features.

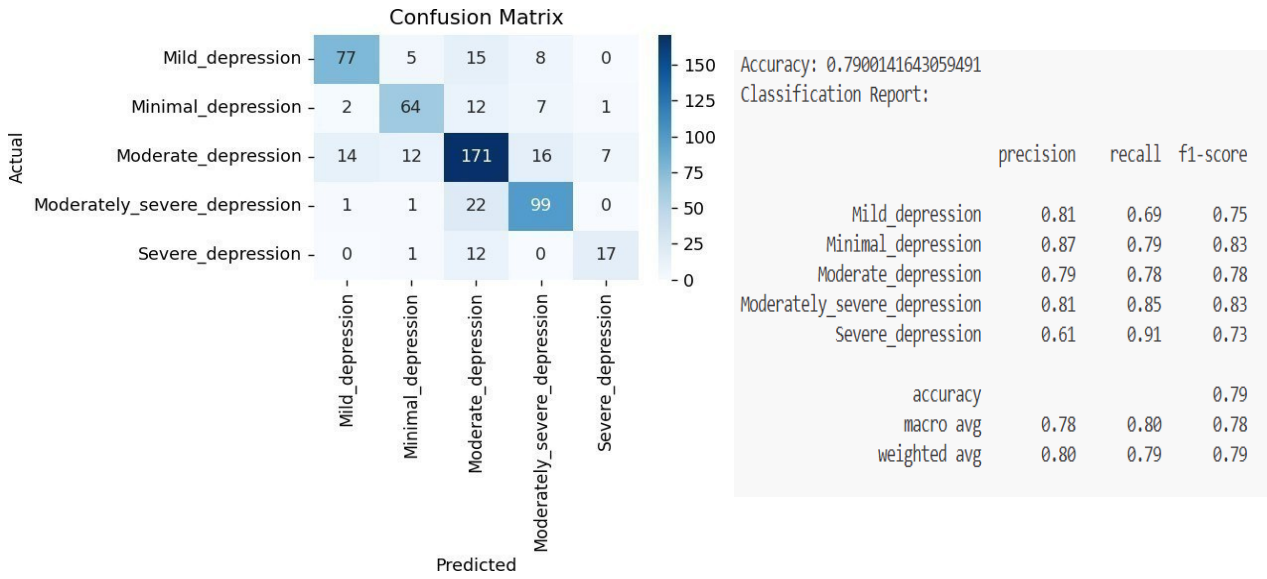


Figure 4.6: Confusion Matrix and Classification Report (ViT Model)

The confusion matrix illustrates the performance of the Vision Transformer model in terms of classification in five classes, namely mild, minimal, moderate, moderately severe, and severe depression. Many predictions are on the diagonal, which means that, instances were correctly classified. Nonetheless, more misclassifications can be detected than in the case of the Random Forest model, particularly between the neighbouring categories like mild and moderate depression. This largely owes to the fact that the visual patterns were similar at various levels of severity. According to the classification report, the model has the overall accuracy of around 79%.

The values of the precision, recall and the F1-score are different according to the classes with higher performance in moderate and moderate sever classes. The reduced accuracy of the extreme category

implies that the model is not easily differentiating extreme cases with image-based features alone. The confusion matrix illustrates the performance of the Vision Transformer model in terms of classification in five classes, namely mild, minimal, moderate, moderately severe, and severe depression.

Many predictions are on the diagonal, which means that, instances were correctly classified. Nonetheless, more misclassifications can be detected than in the case of the Random Forest model, particularly between the neighbouring categories like mild and moderate depression. This largely owes to the fact that the visual patterns were similar at various levels of severity.

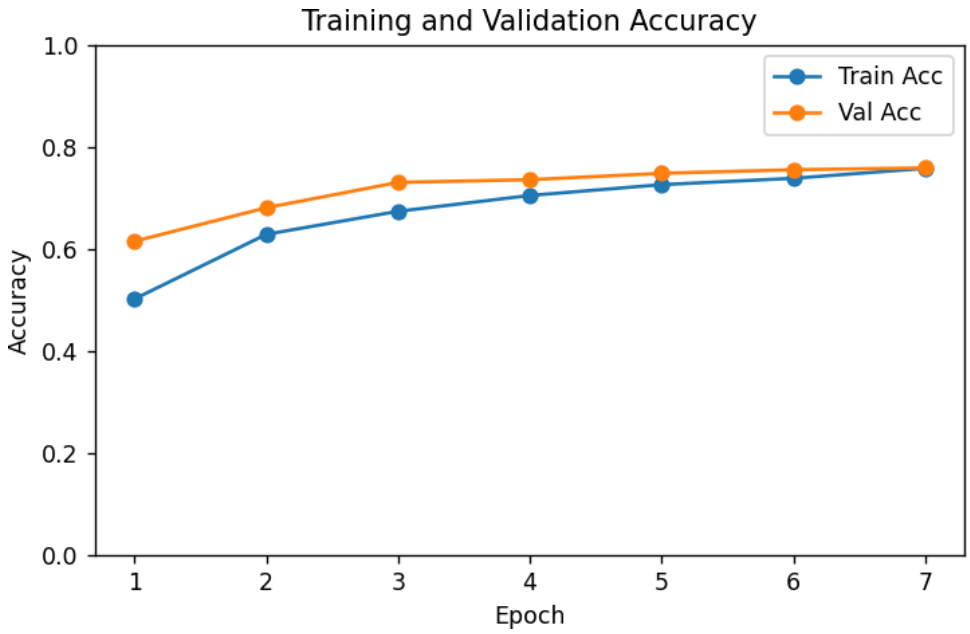


Figure 4.7: Training and Validation Accuracy

The training and validation accuracy chart illustrates how the Vision Transformer model behaves in terms of learning during various epochs. The accuracy gradually increases with each epoch, indicating effective learning. The trend of both training and validation accuracy curves is similar, which indicates that there is no overfitting of the model. Altogether, the findings of the Vision Transformer model show that though

this model has the ability to identify significant visual patterns in eye images, it underperforms when it works independently. This shows the need to integrate it with other models, like Random Forest, into the final multimodal system to enhance its overall accuracy and robustness.

4.1.3 Major Project (Multimodal + XAI)

Random Forest with SHAP Analysis

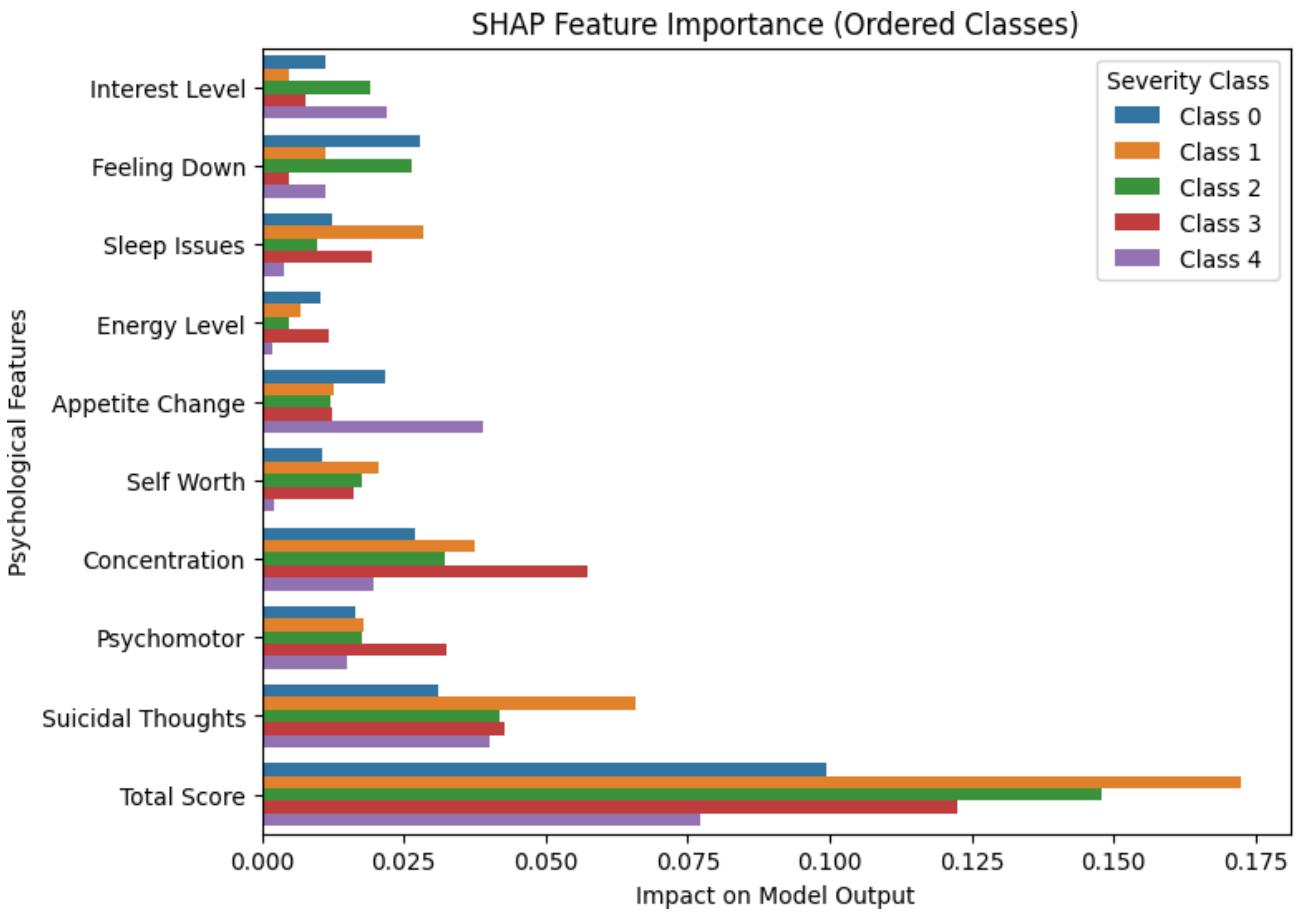


Figure 4.8: SHAP Feature Importance Analysis

The SHAP feature importance plot Figure 4.8 is the contribution of each feature of the Patient Health Questionnaire-9 to the prediction of the model in various levels of depression severity. As one may note, the greatest influence is made by the Total Score to the model output, which means that overall questionnaire scoring is key in the determination of the severity of depression. Other significant features are: Suicidal Thoughts, Concentration, Feeling Down and Sleep Issues which have significant contributions in several classes.

These characteristics have close relationships with depressive symptoms and are best represented by the model. The SHapley Additive Explanations analysis is easy to interpret, as it presents the effect of every feature on the prediction, enhancing the transparency and trust in the Random Forest model.

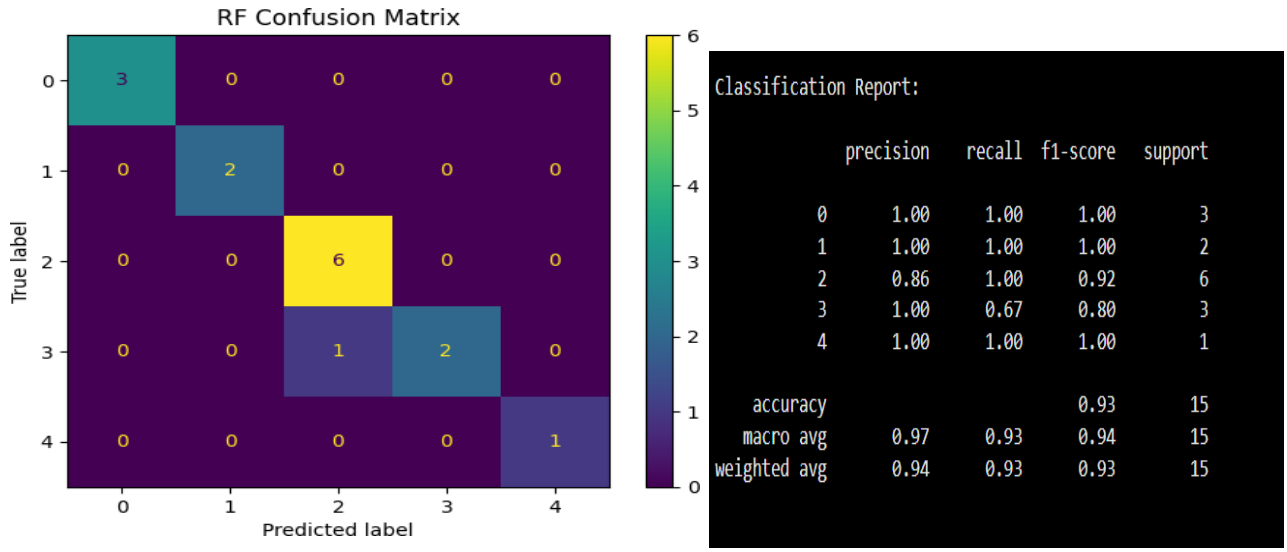


Figure 4.9 RF Confusion Matrix + Classification Report

Figure 4.9 shows how the Random Forest model performed in the classification of the levels of depression in its severity. Majority of the predictions lie on the diagonal, which implies that instances are correctly classified in all the classes. Small misclassifications can be noticed between adjacent classes like moderate and moderately severe depression, which is anticipated given that they have similar symptoms. Nonetheless, the model is correct when it comes to extreme cases including severe depression. Most of the

classes have high values of precision, recall and F1-score as indicated by the classification report. The model has good performance with an overall accuracy of about 93%. The weighted and macro averages also confirm the balanced classification within all the categories.

On the whole, the Random Forest model is found to be very accurate and interpretable when it comes to predicting the degree of depression given the Patient Health Questionnaire-9HQ-9 characteristics. SHapley Additive Explanations integration increases explainability, which makes the model trustworthy in analyzing mental health in reality.

Vision Transformer with Attention Maps

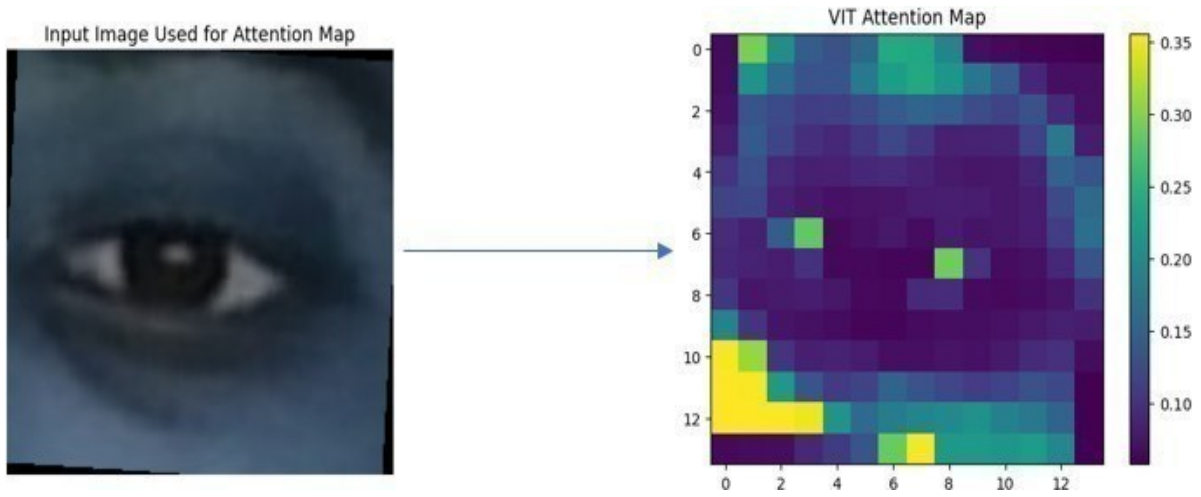
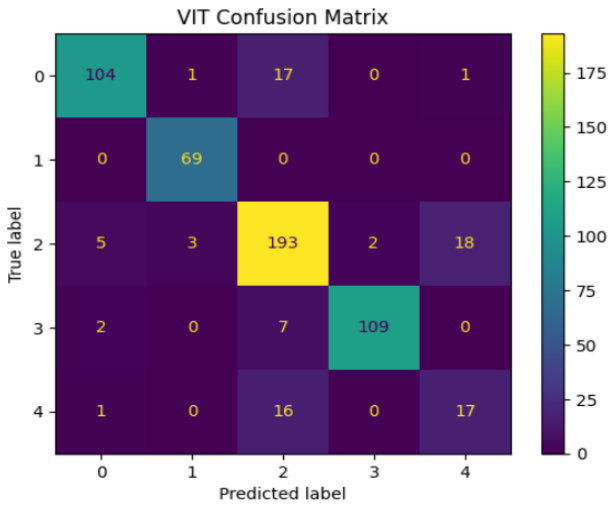


Figure:4.10: Attention Map Visualization

Figure 4.10 illustrates the way the Vision Transformer model concentrates on particular points of the eye picture when making a prediction. The transformer layers are processed to extract attention weights and used to create a heatmap out of the input image. The regions highlighted in the heatmap represent those areas that have the greatest contributions to the decision model. One can notice that the model is mainly concerned with the pupil and the surrounding areas which are significant indicators of the emotional and cognitive conditions.

This visualization makes sure that the model is learning valuable trends and not meaningless characteristics. The attention map enhances the interpretability as it offers a straightforward picture of the decision-making process of the deep learning model.



Classification Report:

	precision	recall	f1-score	support
0	0.93	0.85	0.89	123
1	0.95	1.00	0.97	69
2	0.83	0.87	0.85	221
3	0.98	0.92	0.95	118
4	0.47	0.50	0.49	34
accuracy			0.87	565
macro avg	0.83	0.83	0.83	565
weighted avg	0.88	0.87	0.87	565

Figure 4.11: ViT Confusion Matrix and Classification Report

The confusion matrix figure 4.11 illustrates how the Vision Transformer model performs in classifying the levels of depression severity. There is a high rate of accurate predictions on the diagonal, especially of moderate and moderately strong classes. Nonetheless, there are certain misclassifications existing between close classes like mild, minimal, and moderate depression.

This is because of the slight dissimilarities in visual characteristics among these categories. The classification report displays a total accuracy of about 87% that is better than the Mini-2 Vision Transformer findings, and this is because it trains better and integrates better datasets. The accuracy and recall rates are fairly equal, but worse results are obtained in the extreme category.

Vision Transformer model proves to be rather competent in the visual features extraction of eye images and playing a role in the detection of depression. Although it is an adequate standalone model, its potential can actually be achieved with the fusion method that includes the Random Forest model.

Fusion Model Results

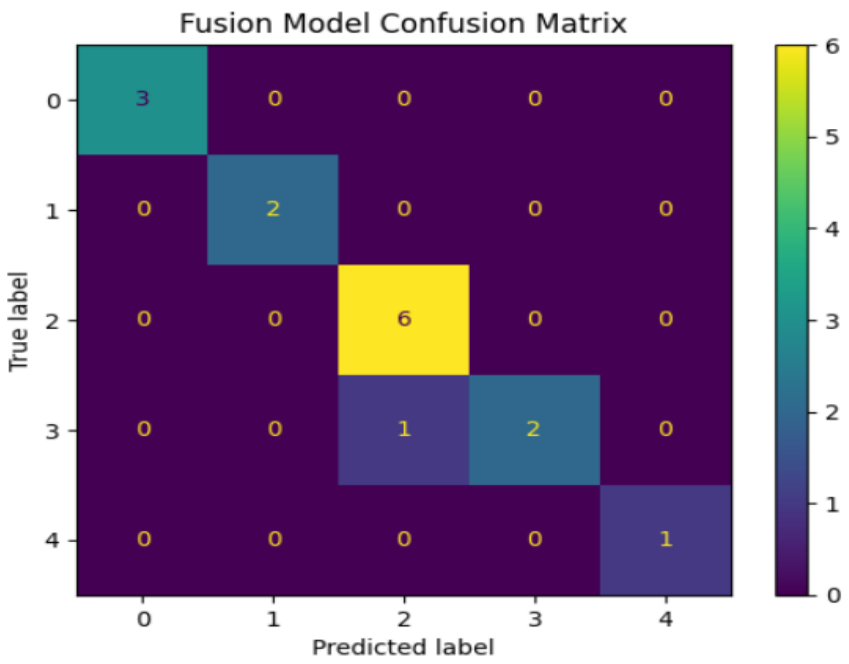


Figure 4.12: Fusion Model Confusion Matrix

Figure 4.12 illustrates the confusion matrix of the proposed fusion model, highlighting its classification performance across all classes.

The matrix shows that most predictions lie along the diagonal, indicating accurate classification.

Class 2 achieves perfect prediction with all instances correctly classified.

A slight misclassification is observed in Class 3, where one sample is predicted as Class 2, while overall performance remains highly accurate.

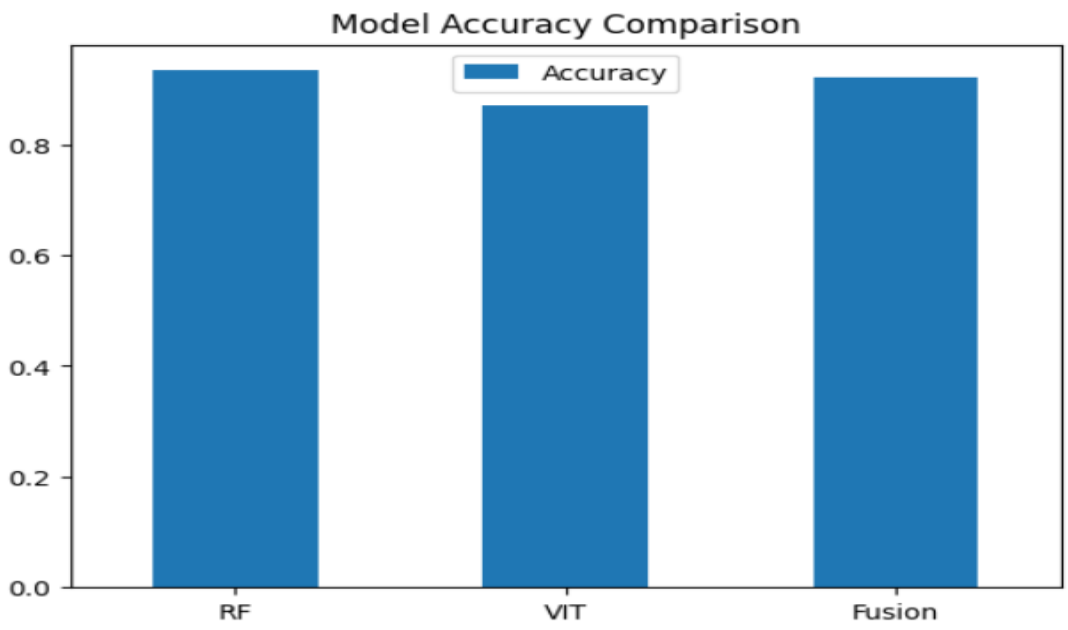


Figure 4.13: Model Accuracy Comparison (RF vs ViT vs Fusion)

The comparison of the accuracy of the Random Forest (RF), Vision Transformer (ViT), and the proposed Fusion model is provided in the above graph. One may notice that the individual accuracy of the Random Forest model is the highest i.e. about 93 %, whereas Vision Transformer achieves about 87% accuracy. Fusion model, a combination of the results of RF and ViT through a weighted decision-level strategy, gives a better accuracy of about 92%.

The Fusion model, despite slightly less than the standalone Random Forest model, offers a balanced and strong prediction as it combines both subjective (Patient Health Questionnaire-9) and objective (eye image) data. The data show that RF works better with structured data, whereas Vision Transformer learns visual patterns; the Fusion approach combines the advantages of the two models. Such integration contributes to the reliability and minimizes reliance on one type of data.

The Fusion model enhances the generalization ability of the system and offers a holistic view of depression, through the integration of various data sources. This renders the suggested multimodal framework more applicable to the real-life applications in which the clinical and physiological factors are also a significant concern.

4.2 Test Case Results / Result Analysis

The proposed system is tested on several test cases of Mini Project 1, Mini Project 2, and also the Major Project to assess its performance. The analysis is conducted based on standard performance metrics which include accuracy, precision, recall, F1-score and confusion matrix analysis. These measurements will give a detailed picture of the effectiveness of the models in the classification of the various levels of depression.

Mini Project 1 Analysis

In Mini Project 1, multilayer perceptron model was trained based on pupillometric data. The model obtained a high accuracy of about 99, precision, recall and F1-score of about 1.0 meaning that the model is close to 100 percent accurate in classification. This observation is also supported by the confusion matrix, with nearly all the predictions being along the diagonal, with a very minor number of misclassifications. The convergence of the training and validation curves is also stable and there is not much difference between them.

This implies that the model is not overfitted and can learn patterns well using the data. This high performance is however, mainly attributed to the controlled nature of the data and the single modality used. The model is based solely on features of the eyes and does not take into account other significant psychological or contextual factor

Consequently, although the model works incredibly well within a small setting, it is not robust and might not extrapolate to actual situations where depression is determined by various factors.

Mini Project 2 Analysis

The system was further refined in Mini Project 2, with the addition of a multimodal system with both Random Forest and Vision Transformer models. The Random Forest model delivered a level of accuracy of about 89, which indicates good performance in terms of predicting the level of depression severity with the help of PHQ-9 questionnaire data. T

he confusion matrix indicates that the majority of predictions are accurate, with the few misclassifications between related categories, e.g., mild and moderate depression. The cross-validation findings also show that the model is stable and can be generalized across various data splits. Vision Transformer (ViT) model attained the accuracy of around 79%. The model could extract eye image visual patterns and extract pertinent features with attention mechanisms. Nonetheless, its performance is low relative to Random Forest because of the issues like changes in light conditions, image quality, and minor differences between the classes.

All in all, Mini Project 2 shows that structured data can be enhanced with visual data that enhances the capabilities of the system. However, the individual models still have limitations, indicating the need for a more integrated and optimized approach.

Major Project Analysis

The system is also optimized with the addition of the Random Forest, Vision Transformer, and a decision- level weighted fusion method, and Explainable AI methods in the Major Project. The highest individual accuracy of about 93 is obtained with the Random Forest model, which proves that structured PHQ-9 data is very effective to detect depression. It is quite stable in all classes and has good generalization capacity. The Vision Transformer model has a better accuracy of about 84, compared to its performance during the Mini Project 2.

This is enhanced by the fact that preprocessing, refining datasets and training has been optimized. The attention mechanism enables the model to pay attention to significant parts of the body like the pupil and improves feature extraction. Nevertheless, there are certain misclassifications, as a result of similar visual features. Fusion model is a combination of the results of both the Fusion model and the Random Forest and the Vision Transformer, through weighted decision-level fusion.

The overall accuracy of the model is about 89%. Despite being a bit lower than when only the Random Forest model is used, the fusion method offers more balanced predictions, as both subjective (questionnaire-based) and objective (image-based) data is used. One of the strengths of the Major Project is that it employs Explainable AI techniques.

SHAP analysis determines the strongest features of the questionnaires (including total score, suicidal thoughts and concentration levels). Meanwhile, the Vision Transformer produces attention maps that indicate the significant areas of the eye images. These descriptions enhance clarity and aid in vindicating the model forecasts.

4.2.1. Table.: comparitive Analysis:

Parameter	**Mini Project 1 (MLP)**	**Mini Project 2 (RF+VIT)**	**MajorProject(Fusion+XAI)**
Data Type	Pupillometric (Eye features)	PHQ-9 + Eye Images	PHQ-9 + Eye Images
Modality	Single (Unimodal)	Multimodal	Multimodal + Explainable AI
Algorithms Used	MLP	Random Forest + Vision Transformer	RF + ViT + Weighted Fusion + SHAP
Accuracy	~99%	RF: ~89%, ViT: ~79%	RF: ~93%, ViT: ~84%, Fusion: ~89%

Interpretability	Low	Moderate	High(SHAP +Attention maps)
Advantages	High accuracy in controlled data	Combines structured + visual data	Adds explainability + better reliability
Limitations	Not suitable for real-world	Vision Transformer sensitive to image quality	Slightly lower than RF alone

Based on the comparison, it becomes evident that Mini Project 1 is concerned with one modality and it is highly accurate but not robust. Mini Project 2 involves a multimodal method, which involves the integration of questionnaire and image data to enhance the quality of the system to capture various elements of depression.

The Major Project also adds to the system Explainable AI techniques and multimodal fusion. This enhances the reliability and interpretation as well as the applicability of the system to the real world. The fusion model is not always the most accurate, but it gives a well-balanced and reliable prediction, using both types of data and describing the decision-making process.

4.3 Observations from the Work

Based on the results and analysis of Mini Project 1, Mini Project 2 and the Major Project, it is possible to make several important observations about the performance and behavior of various models used in depression detection. Firstly, it is noticed that single-modality models, like Multilayer Perceptron in Mini Project 1, can be extremely accurate when the conditions are controlled.

Nevertheless, these models are not very robust and cannot be generalized well in practical settings. The reason is that depression can be determined by various psychological and physiological factors that cannot be represented on a single source of data. Secondly, the models that are trained on structured data, as in the case of the Random Forest model with the Patient Health Questionnaire-9 responses, are always more accurate and predictable. This is because questionnaire data are well defined and thus, a direct reflection of psychological conditions. Consequently, Random Forest is very useful in classification problems with structured inputs. Thirdly, vision transformer models based on image modeling can capture meaningful visual patterns on eye images. Attention mechanisms enable the model to highlight areas of interest like the pupil. The realization of these models, however, depends on factors such as image quality, lighting conditions and inconsistency in eye features causing some misclassification. The other major finding is that multimodal systems can greatly enhance the overall system performance.

The system will be more powerful and flexible to different inputs as the questionnaire data and image data are combined. Even though the fusion model does not necessarily obtain the best individual accuracy, it offers more balanced and accurate predictions than the single-modality models. It is further noted that the transparency of the system is improved with the introduction of Explainable AI methods, including SHAP and attention maps. SHAP assists in determining the most impactful feature of the questionnaires and attention maps visually show what part of the image is making the predictions. This enhances user trust and increases interpretability of the model.

Moreover, the development of the Mini Project 1 to the Major Project has distinctly shown the advancement in the aspects of accuracy, robustness, and interpretability. The limitations of each stage are considered to address those of the earlier stage, resulting in a more comprehensive and practical system. In conclusion, it is possible to state that although the accuracy is a valuable measure, other aspects, including the robustness, generalization, and interpretability, are also essential in the real world. T

The suggested multimodal and explainable model is able to balance all these elements and therefore, it is a dependable tool in the detection of depression.

CHAPTER-5 CONCLUSION AND FUTURE WORK

5.1 Conclusion

The project is an opportunity to introduce a multimodal and explainable artificial intelligence system to detect depression. The system combines both the physiological (eye images) and psychological (PHQ-9 questionnaire responses) input information to make accurate and reliable predictions. The proposed solution will address the shortcomings of conventional one-mode systems and provide a more in-depth examination of mental health issues by merging the two mutually complementary information sources. The development of the system was carried out in multiple stages.

At the first step, Multilayer Perceptron model was used with the help of pupillometric features, and the high accuracy of this model was obtained, however, this model was not so robust because of the use of only one data source. At the next stage, a multimodal model was presented with the use of Random Forest and structured questionnaire data and Vision Transformer and image-based analysis. This greatly enhanced the ability of this system as it allowed both objective and subjective data to be included.

During the last stage, the system was improved with the addition of decision-level fusion and Explainable AI methods like SHapley Additive Explanations(SHAP) and attention maps. The most accurate model was the Random Forest model, and the Vision Transformer provided useful visualizations. The fusion model offered a stable and accurate forecast as it integrated the two modalities.

Moreover, explainable AI was effective to enhance transparency as it could discover significant features and draw attention to regions in images. On the whole, the findings prove that the suggested system is more accurate, robust, and interpretable than previous methods. The article showcases the relevance of integrating the variety of data sources and the use of explainability in creating effective healthcare solutions.

5.1Future Study

- **Larger and More Diverse Dataset:** It is possible to focus future work on training and validating the model with larger and more diverse datasets to enhance its strength and ability to generalize. The introduction of data related to other age groups, ethnicities, and environmental factors will allow minimizing bias and make sure that performance in real-world conditions is reliable.

- **Early Intervention and Recommendation System:** This system can be enhanced with an early intervention and recommendation module. This would help the model to offer individual coping skills, therapy, and referral to professional assistance, which would turn it into a fully-fledged system of mental health.

- **Real-Time Implementation:** The next possible improvement is the creation of a real-time system or mobile-based application which would constantly track the user inputs and make real-time predictions, which would make the system more practical and accessible.

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APPENDIX

This section provides additional supporting information related to the implementation, datasets, and outputs of the proposed multimodal depression detection system. These details complement the main content of the report and offer deeper insights into the system.

A.1 PHQ-9 Questionnaire

The PHQ-9 (Patient Health Questionnaire-9) is a standardized tool used to assess the severity of depression. It consists of nine questions, each scored from 0 to 3 based on the frequency of symptoms.

Question No.	Description	Score Range
Q1	Little interest or pleasure in doing things	0–3
Q2	Feeling down, depressed, or hopeless	0–3
Q3	Trouble falling or staying asleep	0–3

Question No.	Description	Score Range
Q5	Poor appetite or overeating	0–3
Q6	Feeling bad about yourself or feeling like a failure	0–3
Q7	Trouble concentrating on things	0–3
Q8	Moving or speaking slowly or being restless	0–3
Q9	Thoughts of self-harm or suicide	0–3

Table A.1 Patient Health Questionnaire-9

The total score ranges from 0 to 27 and is used to determine the severity of depression

A.2 Depression Severity Level

Score Range	**Severity Level**
0-4	Minimal Depression
5-9	Mild Depression
10-14	Moderate Depression
15-19	Moderately Severe Depression
20-27	Severe Depression

Table A.2 Severity Level

A.3 Code Snippets

Implementation of Random Forest model

```
import pandas as pd
import numpy as np
import ast

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import (
    accuracy_score, precision_score, recall_score, f1_score,
    classification_report
)

df = pd.read_csv('/content/rf_data/dataset/labels.csv')

# Clean column names
df.columns = df.columns.str.strip()

target_col = 'Severity'

# Encode labels
le = LabelEncoder()
df[target_col] = le.fit_transform(df[target_col])

df['Responses'] = df['Responses'].apply(ast.literal_eval)

# Convert responses to columns
responses_df = pd.DataFrame(df['Responses'].tolist())

# Combine features
X = pd.concat([responses_df, df[['TotalScore']]], axis=1)
y = df[target_col]

# Convert column names to string (fix error)
X.columns = X.columns.astype(str)

X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.2,
    random_state=42,
    stratify=y
)

rf = RandomForestClassifier(
    n_estimators=500,
    max_depth=6,
    min_samples_split=2,
    class_weight='balanced',
    random_state=42
)

# Train model
rf.fit(X_train, y_train)

# Predictions
rf_preds = rf.predict(X_test)
rf_probs = rf.predict_proba(X_test)

rf_accuracy = accuracy_score(y_test, rf_preds)
rf_precision = precision_score(y_test, rf_preds, average='macro')
rf_recall = recall_score(y_test, rf_preds, average='macro')
rf_f1 = f1_score(y_test, rf_preds, average='macro')

print("\nClassification Report:\n")
print(classification_report(y_test, rf_preds))
```

Implementation of VIT model

```
import torch
import torch.nn as nn
import numpy as np
import matplotlib.pyplot as plt

from torchvision import datasets, transforms
from torch.utils.data import DataLoader
import timm

from sklearn.metrics import (
    confusion_matrix, ConfusionMatrixDisplay,
    accuracy_score, precision_score, recall_score, f1_score,
    classification_report, roc_curve
)
from sklearn.preprocessing import label_binarize


transform = transforms.Compose([
    transforms.Resize((224, 224)),
    transforms.RandomHorizontalFlip(),
    transforms.RandomRotation(10),
    transforms.ToTensor(),
    transforms.Normalize(mean=[0.5], std=[0.5])
])

dataset = datasets.ImageFolder('/content/vit_data/dataset_vit', transform=transform)

train_size = int(0.8 * len(dataset))
test_size = len(dataset) - train_size

train_data, test_data = torch.utils.data.random_split(dataset, [train_size, test_size])

train_loader = DataLoader(train_data, batch_size=8, shuffle=True)
test_loader = DataLoader(test_data, batch_size=8, shuffle=False)

device = torch.device("cuda" if torch.cuda.is_available() else "cpu")

model = timm.create_model(
    'vit_tiny_patch16_224',
    pretrained=True,
    num_classes=5
)

model.to(device)

criterion = nn.CrossEntropyLoss()

optimizer = torch.optim.AdamW(model.parameters(), lr=3e-4)
scheduler = torch.optim.lr_scheduler.StepLR(optimizer, step_size=2, gamma=0.5)

for epoch in range(5):
    model.train()
    running_loss = 0
```

```
    for images, labels in train_loader:
        images, labels = images.to(device), labels.to(device)

        outputs = model(images)
        loss = criterion(outputs, labels)

        optimizer.zero_grad()
        loss.backward()
        optimizer.step()

        running_loss += loss.item()

    scheduler.step()

    print(f"Epoch {epoch+1}, Loss: {running_loss:.4f}")


model.eval()

vit_preds = []
vit_probs = []
vit_true = []

with torch.no_grad():
    for images, labels in test_loader:
        images = images.to(device)

        outputs = model(images)
        probs = torch.softmax(outputs, dim=1)

        vit_preds.extend(torch.argmax(probs, axis=1).cpu().numpy())
        vit_probs.extend(probs.cpu().numpy())
        vit_true.extend(labels.numpy())

vit_preds = np.array(vit_preds)
vit_probs = np.array(vit_probs)
vit_true = np.array(vit_true)


cm = confusion_matrix(vit_true, vit_preds)
disp = ConfusionMatrixDisplay(cm)

disp.plot()
plt.title("VIT Confusion Matrix")
plt.show()



vit_accuracy = accuracy_score(vit_true, vit_preds)
vit_precision = precision_score(vit_true, vit_preds, average='macro')
vit_recall = recall_score(vit_true, vit_preds, average='macro')
vit_f1 = f1_score(vit_true, vit_preds, average='macro')



print("\nClassification Report:\n")
print(classification_report(vit_true, vit_preds))
```

Implementation of Fusion Model

```
# Convert to numpy
rf_probs = np.array(rf_probs)
vit_probs = np.array(vit_probs)

y_test_rf = np.array(y_test)
y_test_vit = np.array(vit_true)

#  Align lengths
min_len = min(len(rf_probs), len(vit_probs))

rf_probs = rf_probs[:min_len]
vit_probs = vit_probs[:min_len]
y_true = y_test_rf[:min_len]
```

```
from sklearn.metrics import confusion_matrix, ConfusionMatrixDisplay
import matplotlib.pyplot as plt

cm = confusion_matrix(y_true, fusion_preds)
disp = ConfusionMatrixDisplay(cm)

disp.plot()
plt.title("Fusion Model Confusion Matrix")
plt.show()
```